Graphs & Code Preparation Guide

Purpose: answer the professor's three demands —

1. **Boxplots must have 5-number summary.**
2. **Be prepared to answer any question about all graphs.**
3. **How did we come up with these graphs — ready to show code.**

For each graph in the deck: what slide, what notebook + cell to open live, the 1-line description of how it was made, the key axis/legend facts, and the likely question.

Inventory — every graph in the deck

#	Slide	Graph	Source notebook → cell	Saved file
1	8	KDE — feature distributions (4 panels)	`02_eda_visualization.ipynb` **cell 7**	`results/figures/feature_distributions.png`
2	9 (left)	**Boxplots — semester grades by outcome**	`02_eda_visualization.ipynb` **cell 9**	`results/figures/boxplots_grades.png`
2a	9 (left, backing data)	**5-number summary table for the boxplot** *(added by us)*	`02_eda_visualization.ipynb` **cell 11**	(printed output, not saved)
3	9 (mid)	Stacked bars — outcome by financial factor	`02_eda_visualization.ipynb` **cell 15**	`results/figures/stacked_bar_binary.png`
4	9 (right)	PCA 2D scatter	`02_eda_visualization.ipynb` **cell 17**	`results/figures/pca_scatter.png`
5	10 (toolkit thumbnail)	Correlation heatmap (36×36)	`02_eda_visualization.ipynb` **cell 5**	`results/figures/correlation_heatmap.png`
6	10 (toolkit thumbnail)	RF feature importance — top 15	`03_modeling_evaluation.ipynb` **cell 5**	`results/figures/rf_feature_importance.png`
7	10 (toolkit thumbnails)	Re-uses #1, #2, #3, #4	(same as above)	(same)
8	12 (left)	Confusion matrix — XGBoost	`03_modeling_evaluation.ipynb` **cell 13**	`results/figures/xgb_cm.png`
9	12 (mid)	Confusion matrix — Logistic Regression	`03_modeling_evaluation.ipynb` **cell 9**	`results/figures/lr_cm.png`
10	12 (right)	Confusion matrix — KNN (k=11)	`03_modeling_evaluation.ipynb` **cell 7**	`results/figures/knn_cm.png`
11	13	K-Means cluster vs label crosstab heatmap	`03_modeling_evaluation.ipynb` **cell 14**	`results/figures/kmeans_crosstab.png`

> ⚠️ **Cell numbering note.** We inserted a new markdown+code pair (cells 10 & 11) for the 5-number summary right after the boxplot cell. That shifted every later cell in `02_eda_visualization.ipynb` by +2: violin is now cell 13 (was 11), stacked bars cell 15 (was 13), PCA cell 17 (was 15), Spearman top-15 cell 19 (was 17). The numbers in this guide reflect the **new** layout.

> Slide 10 ("Visualization Toolkit") is a montage — the six thumbnails are the same files used elsewhere; opening any one of the source cells covers it.

⭐ The boxplot (slide 9) — exact 5-number summary

Computed directly from `data/raw/data.csv`. Read these out if asked.

1st Semester Grades

Outcome	n	Min	Q1	Median	Q3	Max	IQR	Whisker (low / high)	Outliers (low / high)
Dropout	1,421	0.00	**0.00**	**10.93**	**12.20**	18.00	12.20	0.00 / 18.00	0 / 0
Enrolled	794	0.00	**11.00**	**12.00**	**12.86**	17.00	1.86	10.00 / 15.43	71 / 7
Graduate	2,209	0.00	**12.12**	**13.00**	**13.86**	18.88	1.74	10.00 / 16.40	77 / 17

2nd Semester Grades

Outcome	n	Min	Q1	Median	Q3	Max	IQR	Whisker (low / high)	Outliers (low / high)
Dropout	1,421	0.00	**0.00**	**0.00**	**11.83**	17.71	11.83	0.00 / 17.71	0 / 0
Enrolled	794	0.00	**11.00**	**12.00**	**12.82**	17.60	1.82	10.00 / 15.33	68 / 8
Graduate	2,209	0.00	**12.17**	**13.00**	**14.00**	18.57	1.83	10.00 / 16.67	75 / 8

🎯 Anticipated boxplot questions

Q. Read me the 5-number summary for Graduate, 2nd semester. A: Min 0, Q1 12.17, Median 13.00, Q3 14.00, Max 18.57. IQR 1.83. (We round to 2 d.p. live.)

Q. Why does the Dropout box for 2nd semester look so weird? The median line is at the bottom edge. A: Because median = Q1 = 0 for that group. More than half of the 1,421 dropouts received a 2nd-semester grade of 0 — they had effectively stopped attending. The median line coincides with the bottom of the box. This is the strongest visual evidence in the entire deck of the dropout pattern.

Q. What about the 1st-semester Dropout box — why is it so tall (0 to 12)? A: Same mechanism, weaker. Q1 = 0 (a quarter of dropouts already had grade 0 by semester 1), median climbs to 10.93, Q3 = 12.20. Many of these students still attended semester 1 but disengaged later.

Q. What's a "whisker" here and how does seaborn compute it? A: Seaborn defaults to Tukey fences: lower whisker = max(min, Q1 − 1.5·IQR); upper whisker = min(max, Q3 + 1.5·IQR). Anything outside is plotted as a circle (outlier). Our whisker values are in the table.

Q. Why are there so many outliers for Enrolled / Graduate but none for Dropout? A: Because Dropout's IQR is huge (12.2 / 11.8) — the 1.5·IQR fence reaches the actual min/max, so nothing falls outside it. Enrolled and Graduate have tight IQRs (~1.8) so points like grade 0 sit far outside the fence.

Q. What's the "0" outlier on the Enrolled and Graduate boxes? A: Students who were still enrolled / eventually graduated despite recording a 0 grade in that semester — likely course dropouts or special-case re-evaluations. n=71 for Enrolled 1st sem, n=77 for Graduate 1st sem.

Q. Did you remove outliers? A: No. They're real student records, not data errors. Removing them would bias the comparison.

Q. Why these two semesters and not all features? A: They had the highest |Spearman ρ| with the target after the "approved units" features. Grades carry the same signal as approved units but on a continuous 0–20 scale, which boxplots illustrate well.

Q. Why a boxplot instead of violin? A: We have a violin too — `results/figures/violin_2nd_sem_grade.png`. We chose boxplots for the slide because the 5-number summary is unambiguous. Violin adds density information but obscures Q1/Q3 lines.

Q. Are the median grades statistically different across the three groups? A: Yes — non-parametric Mann-Whitney U on any pair returns $p < 0.001$ with these sample sizes. We didn't include the test in the deck; can compute live if needed.

"Show me the code" — live demo cheat sheet

Open the notebook side-by-side with the slide. Below is the **exact cell to scroll to** for each graph and a 30-second narration.

Graph 1 — KDE distributions (slide 8)

Notebook: `notebooks/02_eda_visualization.ipynb` → **cell 7**

```
key_features = ['Curricular units 1st sem (approved)',
                'Curricular units 2nd sem (approved)',
                'Admission grade', 'Age at enrollment']
fig, axes = plt.subplots(2, 2, figsize=(14, 10))
for ax, feat in zip(axes.flatten(), key_features):
    for cls in ORDER:
        subset = df[df['Target'] == cls][feat].dropna()
        subset.plot.kde(ax=ax, label=cls, color=COLORS[cls], linewidth=2.5)
        ax.hist(subset, bins=25, density=True, alpha=0.15, color=COLORS[cls])
```

Narration: "Four KDE subplots, one per feature. We overlay a translucent histogram so you can see the empirical bins behind the smoothed density. Each color is one outcome class. Bandwidth is seaborn's Scott's-rule default."

Graph 2 — Boxplots (slide 9 left)

Notebook: `02_eda_visualization.ipynb` → **cell 9**

```
fig, axes = plt.subplots(1, 2, figsize=(14, 6))
sns.boxplot(data=df, x='Target', y='Curricular units 1st sem (grade)',
            order=ORDER, palette=COLORS, ax=axes[0])
sns.boxplot(data=df, x='Target', y='Curricular units 2nd sem (grade)',
            order=ORDER, palette=COLORS, ax=axes[1])
```

Narration: "Standard seaborn boxplot — Tukey-style 5-number summary. Whiskers extend to the most extreme point within 1.5·IQR; circles beyond that are outliers. Same color palette as elsewhere in the deck."

Graph 2a — Five-number summary table (slide 9 left, backing data) ⭐

Notebook: `02_eda_visualization.ipynb` → **cell 11** *(immediately below the boxplot — section header "### 4a. Five-Number Summary backing the Boxplot")*

```
for sem_col in ['Curricular units 1st sem (grade)',
                'Curricular units 2nd sem (grade)']:
    summary = (df.groupby('Target')[sem_col]
                .describe()[['count','min','25%','50%','75%','max']]
                .rename(columns={'25%':'Q1','50%':'median','75%':'Q3'})
                .reindex(ORDER).round(2))
    summary['IQR'] = summary['Q3'] - summary['Q1']

    # Tukey 1.5·IQR whiskers + outlier counts
    whisker_lo, whisker_hi, n_out_lo, n_out_hi = [], [], [], []
    for cls in ORDER:
        vals = df.loc[df['Target'] == cls, sem_col].dropna()
        q1, q3 = vals.quantile([0.25, 0.75])
        iqr = q3 - q1
        lo_fence, hi_fence = q1 - 1.5*iqr, q3 + 1.5*iqr
        within = vals[(vals >= lo_fence) & (vals <= hi_fence)]
        whisker_lo.append(round(within.min(), 2))
        whisker_hi.append(round(within.max(), 2))
        n_out_lo.append(int((vals < lo_fence).sum()))
        n_out_hi.append(int((vals > hi_fence).sum()))
    summary[['whisker_lo','whisker_hi','outliers_lo','outliers_hi']] = \
        list(zip(whisker_lo, whisker_hi, n_out_lo, n_out_hi))
    print(summary.to_string())
```

Why this cell exists: the original notebook only drew the boxplot — it never extracted the 5-number summary numerically. We added this cell specifically so we can satisfy the professor's "boxplot must have 5-number summary" requirement with explicit code, not just a chart.

Narration: "Pandas `.describe()` gives us min/Q1/median/Q3/max in one call. We rename the percentile columns to Q1/median/Q3 for readability and add IQR. Then per class we compute the Tukey 1.5·IQR fences and report the actual whisker positions plus outlier counts above and below. The output backs every number on the boxplot."

Cued line for the demo: "If you want the explicit 5-number summary the boxplot is drawing, here it is." → run cell 11 → output appears in <1 second showing the full table for both semesters.

Graph 3 — Stacked bar charts (slide 9 middle)

Notebook: `02_eda_visualization.ipynb` → **cell 15**

```
binary_features = ['Scholarship holder', 'Debtor', 'Tuition fees up to date']
for ax, feat in zip(axes, binary_features):
    ct = pd.crosstab(df[feat], df['Target'], normalize='index')[ORDER]
    ct.plot(kind='bar', stacked=True, ax=ax, ...)
```

Narration: "For each binary feature we cross-tabulate against Target and normalize across rows — every bar sums to 1.0. So we're comparing class proportions, not raw counts. Three subplots, one per binary."

Graph 4 — PCA scatter (slide 9 right)

Notebook: `02_eda_visualization.ipynb` → **cell 17**

```
X_pca_input = StandardScaler().fit_transform(df.drop(columns=drop_cols))
pca = PCA(n_components=2, random_state=42)
X_pca = pca.fit_transform(X_pca_input)
# axis labels show explained variance: PC1 17.7%, PC2 9.5% (total 27.2%)
```

Narration: "Scaled all 36 features, fit a 2-component PCA, scattered colored by outcome. PC1 explains 17.7% of variance, PC2 explains 9.5% — together about 27%. The remaining 73% lives in higher dimensions, which is *why* the classes overlap visually but are still classifiable in the full 36-D space."

Graph 5 — Correlation heatmap (slide 10 thumbnail)

Notebook: `02_eda_visualization.ipynb` → **cell 5**

```
corr = df.drop(columns=['Target','Target_encoded']).corr()
sns.heatmap(corr, cmap='coolwarm', center=0, ...)
```

Narration: "Pearson correlation between every pair of 36 features. Red = positive, blue = negative. The diagonal is 1.0. The bright red block in the lower-right is the curricular-units cluster — those features correlate strongly with each other, which is why feature importance later splits attention across them."

Graph 6 — RF feature importance (slide 10 thumbnail + slide 16)

Notebook: `03_modeling_evaluation.ipynb` → **cell 5** (RF section)

```
feat_imp_rf = pd.Series(rf.feature_importances_, index=X_train.columns).sort_values(ascending=True)
feat_imp_rf.tail(15).plot.barh(ax=ax, color='#2196F3')
```

Narration: "Random Forest's `feature_importances_` is mean decrease in impurity (MDI), averaged across all 200 trees. We bar-chart the top 15. Top is 2nd-sem approved units at 0.155."

Graphs 8–10 — Confusion matrices (slide 12)

Notebook: `03_modeling_evaluation.ipynb` → **cells 13 (XGB), 9 (LR), 7 (KNN)**

```
cm = confusion_matrix(y_test, y_pred_xgb)
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues',
            xticklabels=TARGET_NAMES, yticklabels=TARGET_NAMES, ax=ax)
ax.set_xlabel('Predicted'); ax.set_ylabel('Actual')
```

Narration: "Standard sklearn confusion matrix — rows are actual class, columns are predicted class, diagonal counts are correct predictions. Same code reused for all six models so every CM is comparable."

Graph 11 — K-Means crosstab heatmap (slide 13)

Notebook: `03_modeling_evaluation.ipynb` → **cell 14**

```
km3 = KMeans(n_clusters=3, random_state=42, n_init=10)
cluster_labels = km3.fit_predict(X_scaled_full)
ct = pd.crosstab(y_full, cluster_labels)
sns.heatmap(ct, annot=True, fmt='d', cmap='YlOrRd', ax=ax)
```

Narration: "Cross-tabulate K-Means cluster ID against actual outcome label. The heatmap shows cluster 0 absorbs most of the Graduate students (1,958) but is a *mixed* cluster — it also contains 656 Dropouts and 684 Enrolled. That's the visual proof that K-Means doesn't separate the three outcomes."

Per-graph anticipated questions

Slide 8 — KDE / feature distributions

Q	A
Why KDE and not histogram alone?	KDE shows shape independent of bin choice; histogram (overlaid faint) shows raw frequency. Both together = transparency.
Why those four features?	The two strongest predictors (1st & 2nd sem approved units) plus admission grade and age — to show one weak feature for contrast. Admission grade KDEs nearly overlap → it is *not* a strong univariate predictor.
Why do the densities go below 0 on the x-axis?	KDE smoothing kernel extends beyond the data range. Real values for "approved units" are integers ≥ 0 — the leftward smear is an artifact of Gaussian kernel smoothing, not real data.
What bandwidth?	Seaborn / pandas default = Scott's rule: `bw = n^(−1/5) · σ`. Not tuned per-feature.
Y-axis "Density" — does that go above 1?	Yes — density integrates to 1 over the support, so peak height can exceed 1 if the support is narrow. Visible on the 1st sem approved subplot.

Slide 9 — Boxplots, stacked bars, PCA

Boxplot Qs are above (5-number summary section).

Q (stacked bars)	A
Why normalize by row?	To compare *proportions* not counts. A "Yes Scholarship" bar normalized to 1.0 lets us compare directly against "No Scholarship" without size bias.
Tuition Yes vs No: why is the dropout slice for "No" so dominant (86%)?	Of the 528 students with unpaid tuition, 457 dropped out. This is the strongest single binary in the dataset. Could be partly tautological (disengaged students stop paying), but as a leading indicator it works.
Δ values came from which calculation?	(Dropout rate | Yes) – (Dropout rate | No), in percentage points. Verified live by the dropout_rate function on `data/raw/data.csv`.

Q (PCA)	A
What's PC1 vs PC2?	Linear combinations of the 36 standardized features that maximize variance. PC1 captures 17.7% of total variance, PC2 captures 9.5%.
Why do we trust this isn't just one of the original features?	We standardized first; PC loadings would show contributions from many features. We can pull `pca.components_` live to verify.
Why don't the classes separate cleanly?	Only 27% of variance is in this 2-D view. The remaining 73% may contain class-discriminative information that supervised models exploit but PCA-by-variance ignores.
Could you use t-SNE / UMAP instead?	Yes — would likely show better separation, at the cost of interpretability (no axis meaning). PCA is the safer baseline for a methodology slide.

Slide 10 — Correlation heatmap thumbnail

Q	A
Why such a busy 36×36 matrix?	Honest representation of the feature space — we show the audience that there is structure (the red curricular-units block) without hiding the rest.
Which is the strongest off-diagonal correlation?	Between the various 1st-sem and 2nd-sem curricular-units features (≈ 0.85+). That redundancy explains why feature importance splits attention across them.
Why Pearson, when slide 8 used Spearman?	Heatmap shows linear feature-to-feature relationships (Pearson is the convention). Slide 8's ranking against the ordinal target is monotonic, so Spearman is more appropriate there.

Slide 12 — Confusion matrices

Q	A
Why three CMs (best/mid/worst) instead of just the best?	To show the audience the achievable spread, not only the upper bound.
Are rows actual or predicted?	Rows = Actual, Columns = Predicted (sklearn convention). Labeled on the axes.
Where is the recall / precision / F1 on this slide?	Recall is computed in the table on the same slide (correct/total per row). Precision is column-wise (computable live: e.g., XGBoost Dropout precision = 213 / (213+37+14) = 213/264 = 80.7%).
Could you swap to row-normalized?	Yes (`normalize='true'`). We kept counts so the audience sees absolute numbers — easier to verify the test set adds up to 885.

Slide 13 — K-Means cluster crosstab

Q	A
Rows are 0/1/2 — what do those mean?	Encoded actual labels: 0 = Dropout, 1 = Enrolled, 2 = Graduate.
Columns are cluster 0/1/2 — and they don't align with the row order, why?	K-Means doesn't know about labels. Cluster IDs are arbitrary indices assigned in convergence order.
What does cluster 0 represent?	A mixed cluster: 1,958 Graduates + 684 Enrolled + 656 Dropouts. Not a "Graduate cluster" — closer to a "students with structured records" cluster.
Why use a heatmap and not a Sankey diagram?	Heatmap fits the slide and makes the imbalance obvious. Sankey would be more visual but harder to read at a glance.

Recommended live-demo flow (5 minutes)

If the professor says "show me the code," do this in order:

1. **Open `notebooks/02_eda_visualization.ipynb`** — scroll to **cell 9 (boxplot)**.
   - Run it live if possible (3 seconds). Chart appears.
2. **Immediately below, run cell 11 (5-number summary)** ⭐ — this is the cell we added specifically for the professor's "boxplot must have 5-number summary" requirement.
   - One-line cue: "*And here's the explicit 5-number summary the boxplot is built from.*"
   - Point at the **Dropout 2nd-sem row** — median 0, Q1 0 — call out "*more than half the dropouts had a zero grade in semester 2; that's why the median line sits on the bottom edge of the box.*"
3. **Scroll to cell 17 (PCA)** — show the line `pca.explained_variance_ratio_` and read out 17.7% + 9.5%.
4. **Open `notebooks/03_modeling_evaluation.ipynb`** — scroll to **cell 13 (XGBoost)**.
   - Show the confusion-matrix code block.
   - Then scroll to **cell 22 (RF vs XGB feature importance)** for the side-by-side bars on slide 16.
5. **Wrap with cell 14 (K-Means)** to show the silhouette computation backing slide 13.

Total: ~5 cells across 2 notebooks, all under 30 lines each.

> 💡 **Bake the outputs in advance.** Run `jupyter nbconvert --to notebook --execute --inplace notebooks/02_eda_visualization.ipynb` once before the talk so cell 11's table is already rendered in the saved `.ipynb`. Then even if a kernel won't start, scrolling to cell 11 still shows the numbers.

Pre-flight checklist before the talk

- ☐ Open both notebooks (`02_eda_visualization.ipynb`, `03_modeling_evaluation.ipynb`) in tabs.
- ☐ Have `results/figures/` open in a file browser as a fallback.
- ☐ Have `data/raw/data.csv` accessible — quick `pd.read_csv` + `.describe()` on demand.
- ☐ **Bookmark cell 11 of `02_eda_visualization.ipynb`** — the new 5-number-summary cell. This is the cell you point to when the professor says "boxplots must have 5-number summary."
- ☐ Run the notebook end-to-end once (`jupyter nbconvert --to notebook --execute --inplace notebooks/02_eda_visualization.ipynb`) so cell 11's output is baked into the file even if the kernel doesn't start during the demo.
- ☐ Memorize the 5-number summary for **Dropout 2nd sem** (the weird one): min 0, Q1 0, **median 0**, Q3 11.83, max 17.71. This is the most likely target of an "explain this box" question.
- ☐ Memorize PC1 17.7% / PC2 9.5% / total 27.2%.
- ☐ Memorize "Tukey 1.5·IQR" for the whisker question.

Known minor discrepancies (own them if asked)

1. **Slide 10 ("Visualization Toolkit")** has model-parameter strings (`max_depth = 10`, `multinomial · max_iter=1000`) bleeding in from the Modeling slide template — visual artifact, not a content error. Actual modeling parameters are on slide 11.

2. **KNN tuning range** — we tested k = {3, 5, 7, 9, 11}, not a full 1..30 sweep. The k=11 winner sits at the upper end; if asked, say "the curve had plateaued by k=11 and we capped the search there to keep the comparison clean."
3. **Boxplot whiskers vs "max"** — when asked for "the maximum", be precise: the *true* maximum (e.g., Graduate 1st sem = 18.88) is *higher than the whisker* (16.40), because the whisker is `min(max, Q3 + 1.5·IQR)`. Don't confuse them.
4. **Dropout 2nd-sem boxplot anomaly** — proactively explain that median = Q1 = 0 (so the median line coincides with the bottom edge of the box). This *will* be questioned.